

Coherence Calculation Details

Device Coherence

Download past Device Coherence for your device, or any device hosted by a citizen scientist, as shown on the homepage globe and the Live Data map. Explore this data to find time periods when a device was in high coherence. This may indicate coherence among people in the vicinity of that device; at this point it is still cutting-edge research.

To locate the Device Number for a device:

- If you own a device, log in and get the Device Number displayed on your dashboard
- Hover over any device on the homepage globe to display its Device Number
- Zoom in on the Live Data map, then hover over any device to display its Device Number

Here is an example of the kind of data you will be downloading:

device_number	epoch_time_utc	active_seconds	device_coherence	significance
4	1709497679	3540	41.4633	Normal
4	1709497739	3540	3.586	Normal
4	1709497799	3540	33.7863	Normal
4	1709497859	3540	24.4189	Normal
4	1709497979	3540	53.3823	Normal

- **device_number**: ID number for one device (matches its number on the homepage globe and the Live Data map)
- **epoch_time_utc**: Date and time in UNIX timestamp format. This is the second that this coherence was calculated, in UTC timezone (e.g. see epochconverter.com)
- **active_seconds**: Number of seconds in the past hour during which data was uploaded from this device for calculation (occasionally less than 3600 due to outages, etc.)
- **device_coherence**: Smoothed Device Coherence calculated for this device over the hour preceding epoch_time_utc
- **significance**: Statistical significance of this Device Coherence, ranging from Normal to Extreme

To calculate Device Coherence, check out the math in Bancel and Nelson's Journal of Scientific Exploration 2008 paper about GCP 1, where this metric was originally known as Device Variance. In our case, this coherence is measured across 3 out of 4 RNGs inside one device (the 4th one being sequestered for future research).

The data being downloaded is actually smoothed Device Coherence, as shown on the global map, and it is calculated by taking the absolute value of the 1-hour moving sum of the raw Device Coherence. The significance levels are determined using a simulated distribution, taking the absolute value of the 1-hour moving sum of chi squared data. The levels correspond to p-values: Normal: $p > 0.1$, Elevated: $p < 0.1$, High: $p < 0.05$, Very High: $p < 0.01$, Extreme: $p < 0.001$. (The corresponding values at which Device Coherence transitions from one level to the next most significant i.e. Normal to Elevated, Elevated to High, etc. are 243, 288, 378, and 483, respectively.)

Network Coherence

Download past Network Coherence for the global group, city cluster or any other group. Explore this data to find time periods when an RNG network was in high coherence. This may indicate a change in Network Coherence across a large group of people near that network that are focusing attention or emotion on the same event.

Here is an example of the kind of data you will be downloading.

epoch_time_utc	network_coherence	active_devices
1756684800	-0.869	353
1756684801	-0.7332	353
1756684802	-0.9373	353
1756684803	-0.8036	353
1756684804	0.8289	353

- **epoch_time_utc**: Date and time in UNIX timestamp format. This is the second that this coherence was calculated, in UTC timezone
- **network_coherence**: Network Coherence calculated for this group
- **active_devices**: Number of devices (each with 4 RNGs per device) from which data was uploaded for the calculation. This number changes as the network grows and can occasionally shrink due to outages, etc

For math details regarding the Network Coherence data, check out the [FAQs](#). More can be found as well in Bancel and Nelson's Journal of Scientific Exploration 2008 paper about GCP 1, where this metric was originally known as Network Variance (or NetVar).

A good first step to ensure you understand the data is to reproduce one of the plots like those found on the [Event Analysis](#) webpage.

1. Red Curve: Take a cumulative sum of the Network Coherence in the desired time period
2. Blue Envelope: Calculate the 95% confidence interval of chi squared for increasing degrees of freedom - i.e. degrees of freedom at each point corresponds to the number of seconds on the time (x) axis

Note: Here are some dates for which data is missing or corrupted due to technical issues ([view dates](#))